

ET AI HACKATHON 2026

# IRONHAVEN

## Safety Intelligence

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Real-Time Underground Tunnel Safety Monitoring Dashboard

AI-Powered • 23 Languages • Zero Infrastructure • Single HTML File

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AI Agent (Anthropic)

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Version 1.0

**Single File App**

3000+ Lines of Code

## THE PROBLEM

# Underground tunnels are death traps without data.

On July 2, 2026, the Teesta Stage-VI tunnel in Sikkim collapsed, trapping 30+ workers. Rescue teams had no real-time sensor data, no digital communication, and no centralized command system. This tragedy inspired Ironhaven.

**200+**

Active tunnel projects in India

**50–80**

Worker deaths per year

**0**

Digital monitoring systems deployed

**20+**

Languages spoken by tunnel crews

*“Every minute matters underground. The difference between a safe evacuation and a tragedy is seconds of early warning.”*

## OUR SOLUTION

# What is Ironhaven?

A **real-time safety monitoring dashboard** for underground tunnel construction. Continuous environmental monitoring, bi-directional worker–admin communication, and automated danger alerts — all in a **single HTML file** that runs anywhere.



### 9 Live Sensors

Temperature, CO, Methane, Smoke, Humidity, AQI, Vibration, CO2, Light



### 23 Languages

All 22 Scheduled Languages of India + Tibetan (Bhotia)



### Two-Way Comms

Workers report hazards; admins send real-time instructions back



### Auto Alerts

Audio sirens + visual overlays when thresholds are breached



### 12 Workers

Simulated crew with roles, zone tracking, and safety gear checklists



### Zero Infrastructure

No backend, no database, no install. Works offline in any browser.

## HOW IT WORKS

# System Architecture



### SENSORS

9 environmental  
parameters monitored  
every 3 seconds



### AI ENGINE

Gaussian simulation  
+ threshold detection  
+ trend analysis



### ALERTS

Visual overlays +  
audio warnings +  
push to workers

### ■ Admin Dashboard

- 5 × 9 live sensor grid with sparklines
- Configurable danger thresholds
- Real-time worker zone tracking
- Report inbox with reply capability
- Emergency contacts panel
- Alert feed + audio warning system

### ■ Worker Portal

- Native language interface (23 langs)
- Zone status with color indicators
- Safety gear checklist per zone
- Hazard reporting (voice + photo)
- Incoming admin instructions
- Real-time danger alerts overlay

# 9 Environmental Parameters

| Parameter         | Unit  | Range  | Mean $\pm$ Std Dev  | Danger Threshold |
|-------------------|-------|--------|---------------------|------------------|
| Temperature       | °C    | 22–48  | 30.04 $\pm$ 5.02    | 38°C             |
| Carbon Monoxide   | ppm   | 0–92   | 19.96 $\pm$ 14.66   | 35 ppm           |
| Methane           | %LEL  | 0–13   | 2.91 $\pm$ 2.13     | 5% LEL           |
| Smoke Level       | µg/m³ | 0–178  | 36.05 $\pm$ 27.19   | 70 µg/m³         |
| Humidity          | %RH   | 27–96  | 64.89 $\pm$ 9.89    | 85% RH           |
| Air Quality Index | —     | 0–473  | 102.84 $\pm$ 74.45  | 200              |
| Vibration         | g     | 0–0.15 | 0.05 $\pm$ 0.03     | 0.1 g            |
| CO2 Level         | ppm   | 0–1844 | 509.59 $\pm$ 279.89 | 1000 ppm         |
| Light Intensity   | lux   | 9–520  | 251.44 $\pm$ 79.2   | 100 lux          |

■ **SAFE**

All parameters within normal range

■■ **WARNING**

80–100% of danger threshold

■ **DANGER**

Threshold exceeded — Audio siren

## TUNNEL ZONES

# 5 Operational Zones

| Zone | Name            | Workers | Required Equipment                                 |
|------|-----------------|---------|----------------------------------------------------|
| A    | Intake Shaft    | 4 / 8   | Air quality monitor, Dehumidifier                  |
| B    | Main Bore       | 8 / 15  | Gas detector (CO/CH4), Cap lamp, Fire extinguisher |
| C    | Deep Excavation | 6 / 12  | Multi-gas detector, SCSR device, Rock bolt wrench  |
| D    | Tunnel Face     | 5 / 10  | SCSR, Vibration sensor, Suppression system         |
| E    | Ventilation Bay | 3 / 6   | Anemometer, Wet-bulb thermometer                   |

## MANDATORY SAFETY GEAR — ALL ZONES

- ✓ Hard Hat (Class E)
- ✓ Safety Goggles
- ✓ Respirator Mask (N95/P100)
- ✓ Hi-Vis Vest (Class 3)
- ✓ Cut-Resistant Gloves
- ✓ Steel-Toe Boots
- ✓ Ear Protection (NRR 25+)
- ✓ Two-Way Radio (Ch. 7)

## HAZARD REPORTING

# Report → Respond → Resolve

| Report Type          | Description                | Priority  |
|----------------------|----------------------------|-----------|
| ■ Gas Smell          | Unusual gas detected       | ■■ URGENT |
| ■ Structural Crack   | Crack or deformation found | ■■ URGENT |
| ■ Water Leak         | Seepage or flooding        | Standard  |
| ■ Unusual Noise      | Rumbling / abnormal sound  | Standard  |
| ■■ Excessive Heat    | Temperature spike felt     | ■■ URGENT |
| ■■ Equipment Failure | Malfunction detected       | Standard  |
| ■■ Poor Visibility   | Dust or smoke buildup      | Standard  |
| ■ Electrical Hazard  | Sparking / exposed wiring  | ■■ URGENT |
| ■ Other              | Any other abnormality      | Standard  |



### Worker

Files hazard report  
with voice + photo



### Admin Inbox

Red = urgent alert  
Amber = standard



### Response

Acknowledge + send  
instructions to worker

## MULTILINGUAL

# 23 Languages

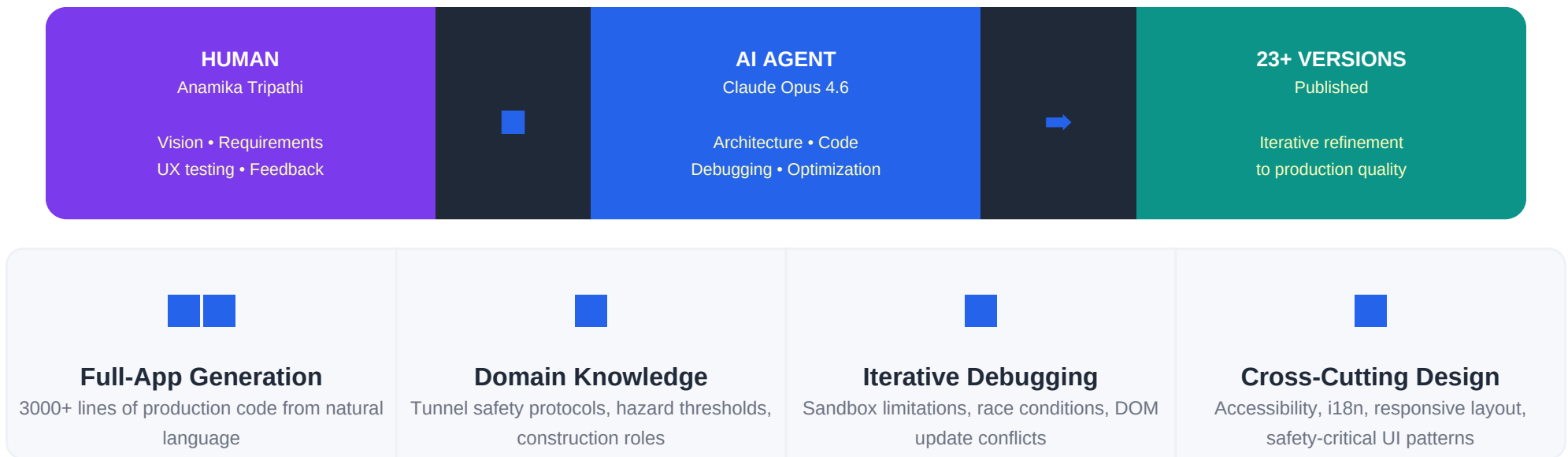
Every Scheduled Language of India + Tibetan. A safety system in English alone excludes the majority of India's migrant tunnel workers.

|                  |                   |                  |                    |                   |
|------------------|-------------------|------------------|--------------------|-------------------|
| English          | Hindi — ■■■■■■    | Bengali — ■■■■■  | Telugu — ■■■■■■    | Tamil — ■■■■■■    |
| Marathi — ■■■■■  | Gujarati — ■■■■■■ | Kannada — ■■■■■■ | Malayalam — ■■■■■■ | Odia — ■■■■■■     |
| Punjabi — ■■■■■■ | Assamese — ■■■■■■ | Urdu — ■■■■      | Maithili — ■■■■■■  | Sanskrit — ■■■■■■ |
| Konkani          | Dogri             | Nepali           | Sindhi             | Meitei (Manipuri) |
| Santali          | Kashmiri          | Tibetan (Bhotia) |                    |                   |

**Every label, alert, and instruction** is delivered in the worker's own script — safety should never be lost in translation.

# Human + AI Collaboration

Ironhaven was created through **human–AI collaboration**. The human defined the vision and tested the experience. The AI agent designed, coded, debugged, and iterated the entire 3000+ line application across 23 published versions.



## CURRENT LIMITATIONS

# Honest Constraints



### Simulated Data

All sensor readings are algorithmic — no real IoT hardware connected



### No Backend

Data lives in browser memory + localStorage; no server persistence



### Single-User

Runs in one browser tab; no multi-device synchronization



### No Authentication

No login system or role-based access control



### Browser Sandbox

confirm(), tel: links, alert() restricted in hosted artifacts



### Local-Only Comms

Worker-admin messaging stays in-app; no external push notifications



### No GPS Tracking

Worker position is zone-based simulation, not real indoor positioning

**These are deliberate scope decisions, not failures.** The prototype proves the concept; the next slide shows how each constraint is resolved in production.

REAL-WORLD DEPLOYMENT

# Production Architecture

| Location      | Infrastructure          | Function                             |
|---------------|-------------------------|--------------------------------------|
| Tunnel Zone   | Edge Gateway + Sensors  | Local aggregation, 5s push           |
| Tunnel Portal | On-Site Server Rack     | Data processing, local DB, WebSocket |
| Surface HQ    | Cloud / On-Prem Cluster | Central dashboard, analytics, ML     |
| Emergency     | API Integration         | Auto-alerts to fire/ambulance/rescue |
| Regulatory    | Compliance Engine       | Automated DGMS / OSHA safety logs    |

RECOMMENDED SENSOR HARDWARE

| Parameter       | Sensor                                |
|-----------------|---------------------------------------|
| Gas Detection   | Honeywell BW Ultra / Drager X-am 8000 |
| Temp / Humidity | Vaisala HMT330 (IP67, tunnel-rated)   |
| Vibration       | PCB Piezotronics 608A11 (triaxial)    |
| Air Quality     | TSI DustTrak / Aeroqual AQM 65        |
| Smoke / Fire    | Xtralis VESDA VLP (aspirating)        |

BACKEND STACK



## FUTURE ROADMAP

### PHASE 1

3–6 months

Real IoT sensor integration via MQTT • User authentication + RBAC • WebSocket real-time multi-device sync • Progressive Web App (offline-first)

### PHASE 2

6–12 months

ML anomaly detection (LSTM/Transformer) • Predictive alerts ("Methane trending → danger in 12 min") • Computer vision PPE compliance from CCTV • NLP hazard report severity classification

### PHASE 3

12–24 months

Multi-tunnel regional command centers • BIM integration for 3D tunnel visualization • Regulatory automation (DGMS, ISO 45001) • Wearable + smart helmet integration

### PHASE 4

24+ months

AI copilot for safety officers • Digital twin with real-time 3D rendering • Federated learning across tunnel sites • Autonomous drone post-blast inspection

# Zero External Libraries

| Category      | Technologies                                                    |
|---------------|-----------------------------------------------------------------|
| Frontend      | HTML5, CSS3 (Custom Properties, Flexbox, Grid), ES6+ JavaScript |
| Charting      | Custom SVG sparkline renderer (zero library dependency)         |
| Audio Alerts  | Web Audio API (OscillatorNode + GainNode)                       |
| Media Capture | MediaRecorder API (voice), FileReader API (photos)              |
| Persistence   | localStorage (thresholds, contacts, language, theme)            |
| Theming       | CSS Custom Properties with dark/light toggle                    |
| Deployment    | Vercel (static hosting, global CDN, auto-SSL)                   |
| AI Agent      | Claude Opus 4.6 (Anthropic) via Claude Code                     |
| Documentation | ReportLab Platypus (Python PDF generation)                      |

## ■ ZERO EXTERNAL JAVASCRIPT LIBRARIES

Charts, animations, audio, internationalization — everything is built with **native browser APIs only**. No npm packages, no CDN dependencies, no supply-chain risk. The entire app loads instantly and works even in bandwidth-constrained tunnel environments with spotty connectivity.

# Safety Intelligence

*“Every tunnel deserves real-time awareness.  
Every worker deserves safety in their own language.  
Every administrator deserves data that saves lives.”*

& **Claude** (Anthropic)

**ET AI Hackathon 2026**

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